MIL-C-39012/38F
14 December 1987
SUPERSEDING
MIL-C-39012/38E
30 September 1982

MILITARY SPECIFICATION SHEET

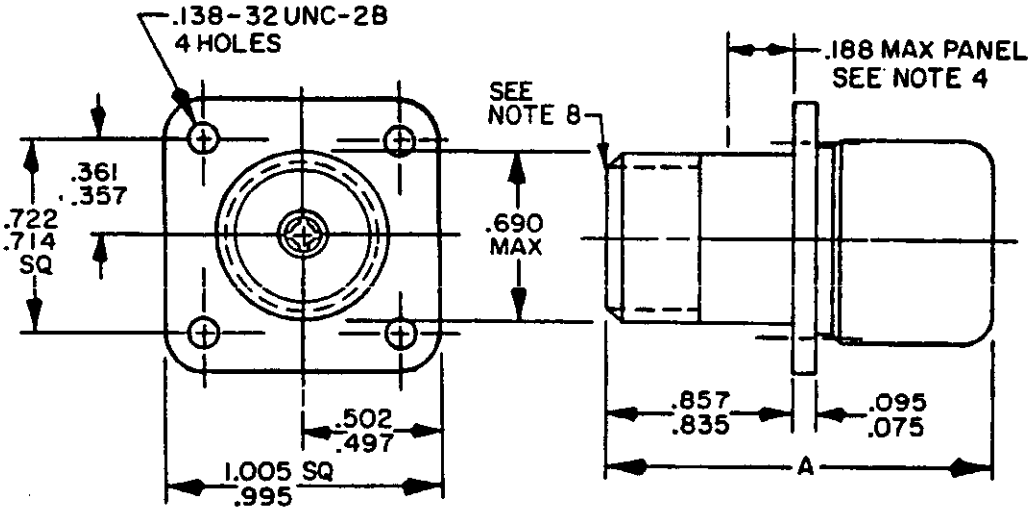
CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY,

(SERIES SC (CABLED), SOCKET CONTACT, FLANGE MOUNTED, REAR MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-39012.

MARKING IMPLEMENTATION DATE FOR CATEGORY B, SEE TABLE IV

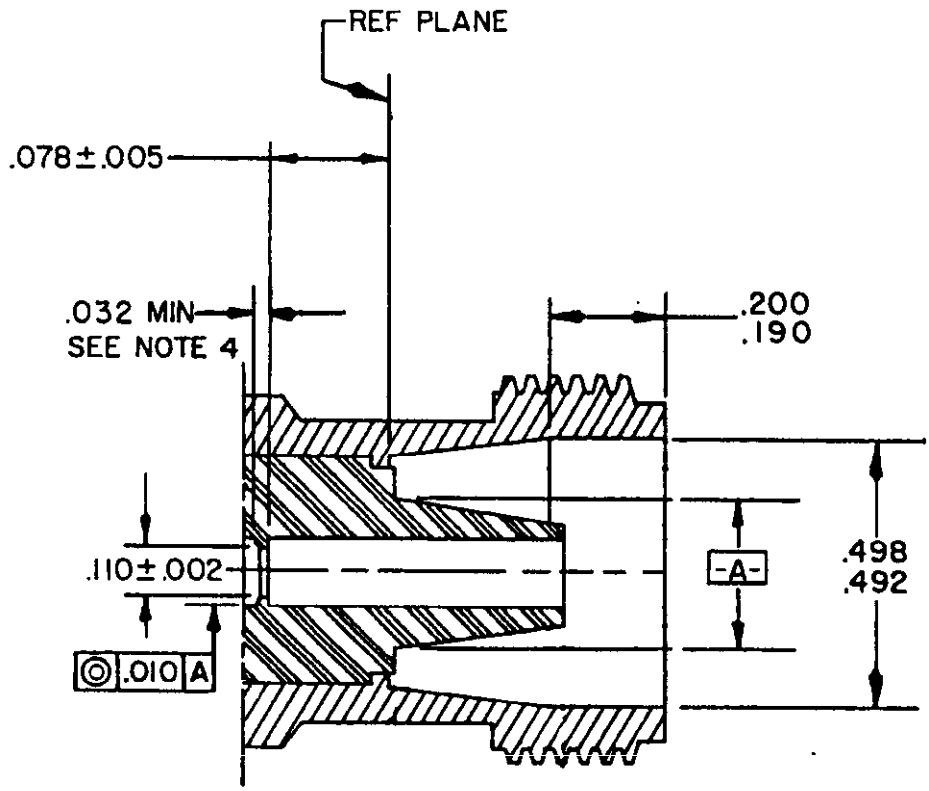


Inches	mm
.075	1.91
.095	2.41
.138	3.51
.188	4.78
.357	9.07
.361	9.17
.497	12.62
.502	12.75
.690	17.53
.714	18.14
.722	18.34
.835	21.21
.857	21.77
.995	25.27
1.005	25.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. For dimension A, see table I and III.
4. Receptacle not recommended for use on panels with greater than .078 (1.98 mm) maximum thickness.
5. Wrench flats to accommodate standard wrench in accordance with H-28, appendix 10.
6. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
7. All undimensioned pictorial configurations are for reference purposes only.
8. Series SC, socket contact interface in accordance with MIL-STD-348, 309.2.

FIGURE 1. General configuration.

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CATEGORY D

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.124	3.15	.309	7.85
.005	0.13	.140	3.56	.374	9.50
.007	0.18	.190	4.83	.411	10.44
.010	0.25	.200	5.08	.415	10.54
.032	0.81	.250	6.35	.482	12.24
.047	1.19	.272	6.91	.491	12.47
.077	1.96	.273	6.93	.495	12.57
.078	1.98	.300	7.62	.498	12.65
.110	2.79	.303	7.70	.630	16.00
.119	3.02			.6875	17.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Chamfer is optional.

FIGURE 2. Mating dimensions for socket termination, category D only.

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TABLE I. Dash numbers, cross-reference, and dimensions.

Dash number 1/	Applicable cable 2/ M17/	Dimensions	Inches (millimeters) Maximum
CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED)			
0010 3/	065-RG165 074-RG213 75-RG214 4/ RG-225/U 5/	A	1.687 (42.85)
0011 3/	073-RG212 4/ 112-RG304		
0012 3/	074-RG215	A	2.500 (63.50)
0013 3/	92-RG115 5/	A	1.687 (42.85)
CATEGORY C - FIELD REPLACEABLE (MIL-C-22520 CRIMP TOOL) 6/ SEE FOOTNOTE NEXT TO APPLICABLE CABLE FOR CRIMP DIE			
0006 3/	065-RG165 5/ 7/ 074-RG213 4/ 7/	A	1.843 (46.81)
0007 3/	75-RG214 4/ 7/ RG-225/U 5/ 7/		
0008 3/	073-RG212 4/ 8/ 112-RG304 8/		
0018 3/	92-RG115 5/ 7/		
CATEGORY D - FIELD REPLACEABLE - DEFINED PIECE PARTS 6/ 9/ 10/			
0501 3/	75-RG214 4/ RG-225/U 5/ 127-RG393	A	1.859 (47.23)
0502 3/	074-RG213 4/		
0503 3/	060-RG142 4/ 5/ 128-RG400		

1/ For cross-reference of dash number to superseded part number or type designation, see table IV.

2/ The latest version of each cable shall be applicable.

3/ These connectors have captivated center contacts.

4/ Cable to be used when performing tests requiring cable except as in 5/.

5/ Cable to be used for the +200°C temperature cycling tests.

6/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.

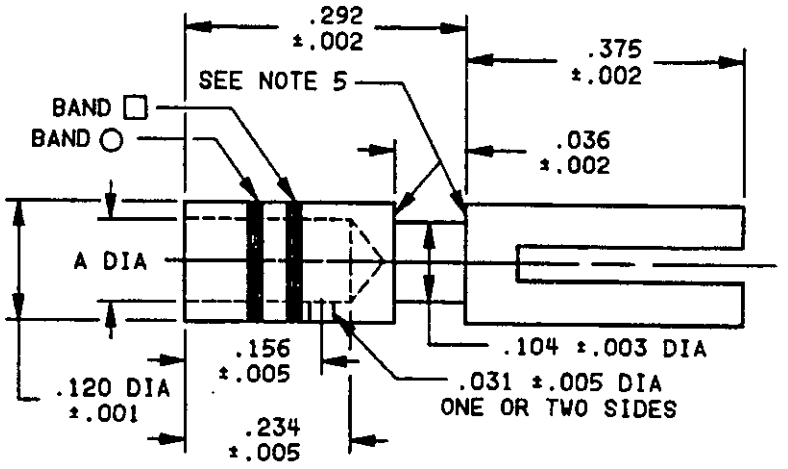
7/ M22520/5-61

8/ M22520/5-35 Closure A or M22520/5-55 Closure A

9/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

10/ Not to be used in Army equipment.

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Inches	mm	Inches	mm
.001	0.03	.234	5.94
.002	0.05	.250	6.35
.003	0.08	.292	7.42
.005	0.13	.351	8.92
.015	0.38	.375	9.52
.031	0.79	.394	10.01
.036	0.91	.406	10.31
.043	1.09	.418	10.62
.098	2.49	.438	11.13
.104	2.64	.492	12.50
.120	3.05	.500	12.70
.156	3.96	.600	15.24
.220	5.59		

CENTER CONTACT

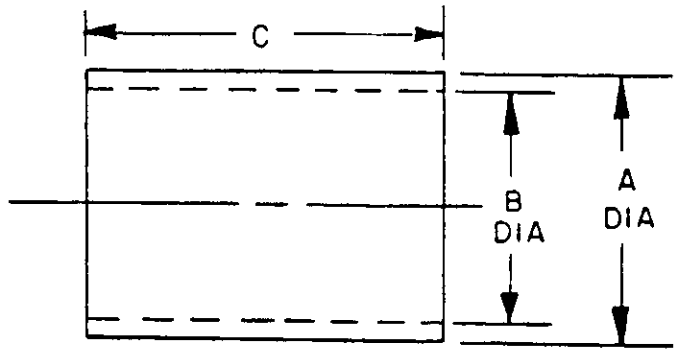
Dash no.	Contact no. 2/	A	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
0501 0502	38-10	.098 ±.002	M22520/1-01	M22520/1-14	60 lbs	Red	White
0503	38-11	.043 +.001 -.002	M22520/1-01	M22520/1-14	20 lbs	Blue	

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Contact material shall be phosphor bronze or copper beryllium.
4. Crimp tensile test shall be in accordance with MIL-C-39029.
5. .003 maximum break.
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FIGURE 3. Contact and ferrule dimensions for category D only.

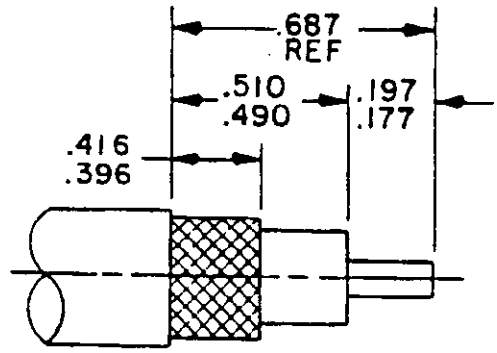
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Dash no.	Ferrule no. 2/	A	B	C ±.015	Basic crimp tool 1/ M22520/5	Crimp die or positioner M22520/5-
0501	38-50	.492 +.003 -.000	.438 +.003 -.000	.600		25 Closure A or 61
0502	38-51	.492 +.003 -.000	.418 +.003 -.000	.600	M22520/5-01	
0503	38-52	.250 ±.001	.220 ±.001	.500		05, 11, 57 Closure A or 19 Closure B

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

FIGURE 3. Contact and ferrule dimensions for category D only - Continued.



Inches	mm
.177	4.50
.197	5.00
.396	10.06
.416	10.57
.490	12.45
.510	12.95
.687	17.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 11,000 MHz.

Voltage rating:

1,000 volts rms maximum working voltage at sea level.

250 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Design and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force - Not applicable.

Torque - 2 inch-pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque - 6 to 10 inch-pounds.

Mating characteristics: See figure 2 and MIL-STD-348 for dimensions.

Center contact (socket):

Oversize test pin - .098 diameter minimum, 16 microinch finish (nonclosed entry contacts only).

Insertion depth - .125 minimum.

Number of insertions - 1.

Insertion force test - Steel test pin diameter .092 minimum.

Test pin finish - 16 microinches.

Insertion force - 2 pounds maximum.

Withdrawal force test: Steel test pin diameter .090 maximum.

Withdrawal force - 2 ounces minimum.

Test pin finish - 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms minimum.

Center contact retention: (Applicable to captivated-center-contact connectors only.) 15 pounds minimum axial force for all cables except RG400 and RG142; 6 pounds minimum for RG400 and RG142.

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Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR):

From 500 to 11,000 MHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower; 1.30 maximum.

Swept frequency VSWR test setup:

Item 6 - VSWR shall be less than 1.015 +.005 F (F in GHz).

Item 16 - VSWR shall be less than 1.015 +.005 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.045 +.019 F (F in GHz).

Group B inspection - VSWR shall be less than 1.10 +.01 F (F in GHz).

Qualification and group C inspection - VSWR shall not exceed 1.15.

Connector durability:

500 cycles minimum at 12 cycles/minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms maximum.

	Initial	After environment
Center contact	1.0	1.5
Outer contact	.15	Not applicable
Braid to body	.05	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 3,000 volts rms at sea level for connectors using other than RG400 and RG142; 1,500 Vrms minimum for connectors using RG400 and RG142.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see table I and III).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Voltage - 750 volts rms minimum.

Altitude - 70,000 feet.

RF high potential withstanding voltage:

Voltage and frequency - 2,500 volts rms tested at a frequency from 5 to 7.5 MHz.

Leakage current - Not applicable.

Cable retention force:

Noncrimp assemblies: 75 pounds minimum.

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Crimp assemblies:

50 lbs minimum for cables .155-.189 OD.

60 lbs minimum for cables .190-.229 OD.

75 lbs minimum for cables .230-.249 OD.

90 lbs minimum for cables .250 OD and larger.

Coupling mechanism retention force: Not applicable.

RF leakage: -90 dB minimum, tested at a frequency between 2 and 3 GHz.

Insertion loss:

.15 dB maximum at 9 GHz.

.05 $\sqrt{F\ (GHz)}$ dB maximum tested at 3 GHz and 6 GHz.

Part number: M39012/38- (dash number from table I or "B" number from table III).

TABLE II. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
	M39012/38	M39012/38
I	-0010	-0010 -0011 -0012 -0013
II	B0015	B0015 B0016 B0017
III	-0006	-0006 -0007 -0008 -0018
IV	-0501 -0502	-0501 -0502 -0503
V	-0503	-0503

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE III. CATEGORY B - NONFIELD REPLACEABLE (SPECIAL TOOLS MAY BE REQUIRED). 1/

[NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY.]

Part no. 2/ M39012/38B	Applicable cable 3/ M17/	Dimensions	Inches (millimeters) maximum
0015 4/	75-RG214 5/ RG-225/U 6/	A	1.843 (46.81)
0016 4/	073-RG212 112-RG304		
0017 4/	92-RG115 6/		

1/ For maintenance replacements for category B, see table V.

2/ For cross-reference of dash number to superseded part number or type designation, see table IV.

3/ The latest version of each cable shall be applicable.

4/ These connectors have captivated center contacts.

5/ Cable to be used when performing tests requiring cable except as in 6/.

6/ Cable to be used for the +200°C temperature cycling tests.

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TABLE IV. Cross-reference of part numbers. 1/

Preferred part number	Substitute for part number or type designation 2/
M39012/38	M39012/38
-0010	-0001
-0011	-0002
-0012	-0009
B0015	-0015
B0016	-0016
B0017	-0017

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The connectors that are in stock or distribution that were previously qualified and marked with the old part number shall also be considered acceptable for Government use until connector stock is purged.

2/ The superseded part number or the type designation is for cross-reference only. Where a superseded part number or type designation is not given, none was assigned or will be assigned. The part number M39012/38-XXXX shall be used in all cases for marking and identifying the connector.

TABLE V. Maintenance replacements for category B.

Category B 1/ number	Category C dash number	Category A dash number	Category D dash number
B0015	0007	0010	0501
B0016	0008	0011	---
B0017	0018	0013	---

1/ Category B connectors are for original installation only. They will not be stocked or procured by the Government.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - EA, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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